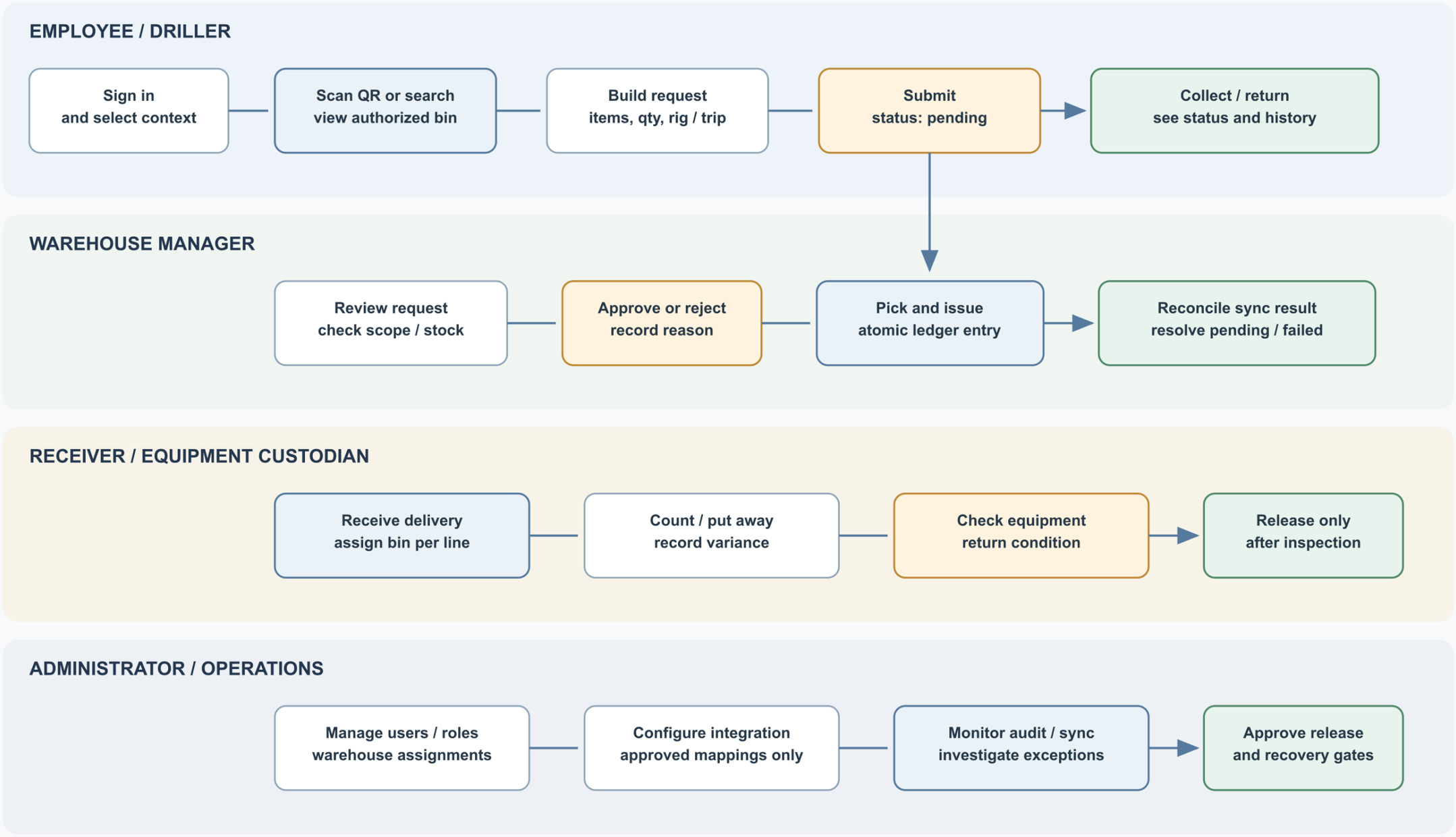
User role flows

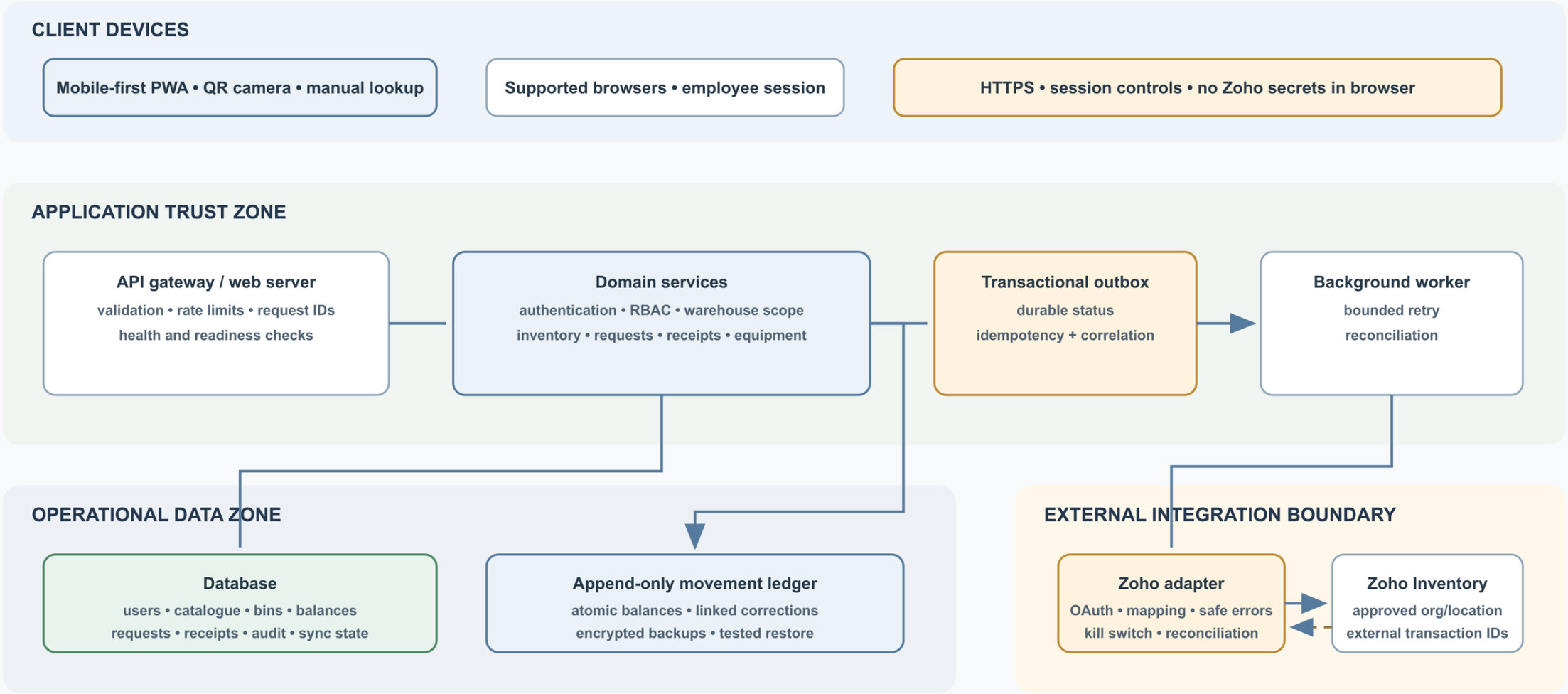
Target operating model • role boundaries must be enforced by the API



All stock changes require server authorization, idempotency, an audit record and an explicit integration outcome.

Production target architecture

Logical design • framework, cloud provider, database and identity vendor require source audit and owner approval



CROSS-CUTTING PRODUCTION CONTROLS

CI/CD • separate environments • least-privilege secrets • logs/metrics/alerts • dependency/security review • incident runbooks • restore/rollback drills

Recommended target capabilities; actual product choices and deployed topology remain to be confirmed from source and hosting configuration.

Solid arrows show request/data flow. Dashed arrow shows provider response. No direct browser-to-Zoho calls.

Stock request, approval and issue

Target state machine • validate access and stock before committing a movement

EMPLOYEE

Create request
items • quantity • rig/trip

→ **Submit once**
idempotency key

→ **Pending review**
visible request status

→ **Collect / see outcome**
issued, rejected or pending

APPLICATION API AND DATABASE TRANSACTION

Authenticate + authorize
role • warehouse • request state

→ **Enough available?**

Yes → **Manager decision**
approve / reject + reason

→ **Atomic issue transaction**
lock balance • write ledger • audit

→ **Commit + outbox**
one movement per key

No: hold/reject + reason

INTEGRATION WORKER

Read durable command
correlation + same idempotency

→ **Send approved Zoho action**
server-side credentials

→ **Known result?**

Yes → **Mark complete**
store external ID

No / timeout → **Unknown / failed**
queue + alert + reconcile

Retry only when safe; otherwise reconcile before replay

INVARIANTS

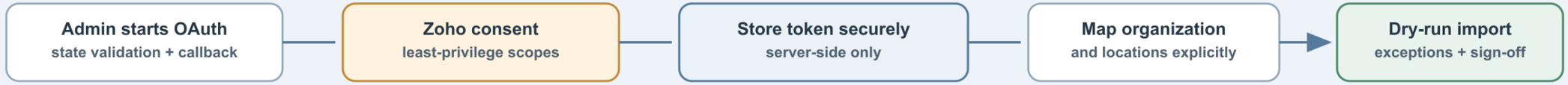
Atomic local ledger • no duplicate retry • no silent external mismatch • correction uses linked event • tests never post to production

Exact local-versus-Zoho commit ordering depends on the approved authority policy and must be confirmed before enabling live writes.

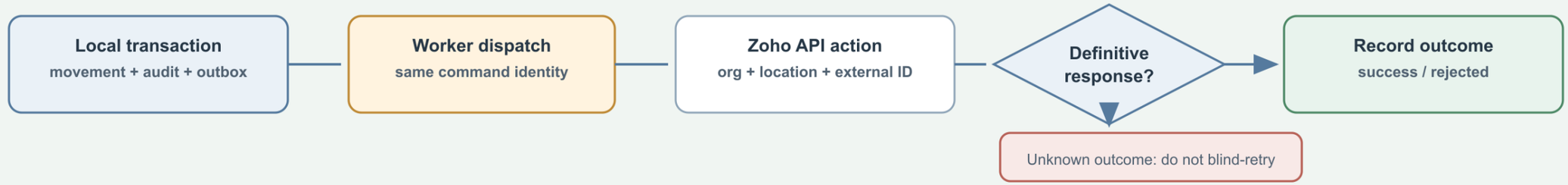
Zoho sync and reconciliation

Target integration workflow • live state, credentials and provider behavior remain unverified

ONE-TIME CONNECTION AND IMPORT



OUTBOUND STOCK COMMAND



RECONCILIATION AND CONTROL



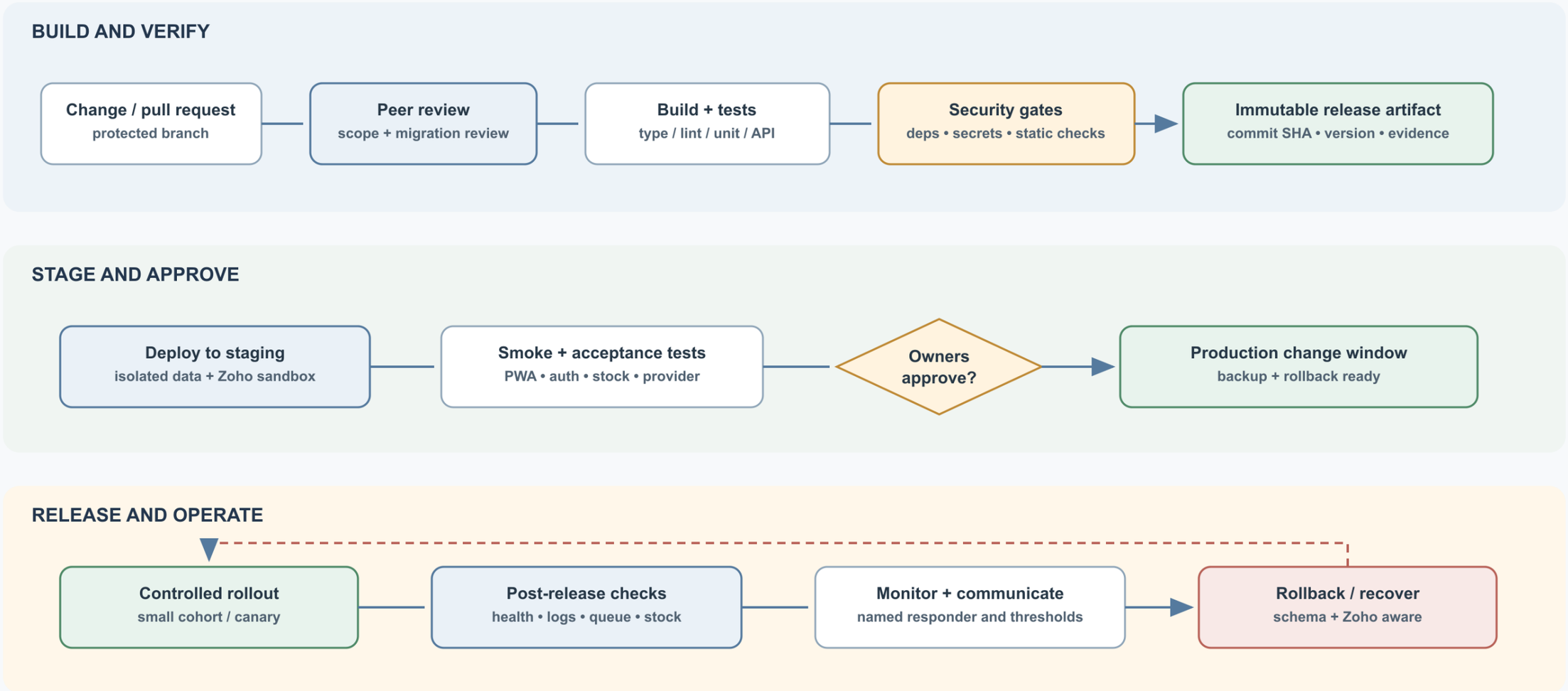
Production gates

Sandbox all mutations • verify OAuth scope and secrets • test token expiry, rate limits, duplicates, timeouts • canary cutover • reconcile before resuming uncertain writes

Business approval is required to define which system owns each data domain and when local versus Zoho state is considered committed.

Production CI/CD release pipeline

Recommended controls • verify the current repository and hosting provider before adoption



Never use production secrets in CI • run tests against live inventory • deploy an unreviewed schema change • blindly retry an unknown Zoho result

Pipeline stages are target practice; exact CI vendor, hosting steps and approval roles are undecided until source and environment audit.

Conceptual data model

Target domain model • entities and cardinalities are conceptual, not a verified schema

Warehouse
id • code • name • status
Zoho location mapping

Bin
id • warehouse_id • code
status • opaque QR token

StockBalance
product_id • bin_id
on_hand • reserved • version

Product
id • SKU • name • base_unit
Zoho item ID

User
identity • role • status
warehouse assignments

Movement
product • from/to bin • qty
kind • actor • source • time
idempotency key • immutable

Request + lines
requester • warehouse • state
product • requested/issued qty
rig / trip reference

Reservation
request line • product • bin
quantity • active/released
release exactly once

Receipt + lines
supplier • reference • receiver
accepted/damaged qty • unit
target bin for each line

SyncCommand
provider • operation • state
correlation • retry • external ID
durable outbox

EquipmentAsset
asset tag • status • condition
home warehouse • unique tag
tracked individually

EquipmentLoan
asset • borrower • due/return
checkout and return condition
one active loan per asset

Trip / RigKit
warehouse • rig • dates • site
requested lines • kit template
planning and reservation source

AuditEvent
actor • action • entity • time
reason • before/after summary
append-only restricted access

RELATIONSHIPS AND INTEGRITY

Warehouse 1→many Bin • Bin + Product 1→1 StockBalance row per pair • Product 1→many Movement and RequestLine

Request 1→many RequestLine • RequestLine 1→many Reservation records • Receipt 1→many ReceiptLine • ReceiptLine creates Movement

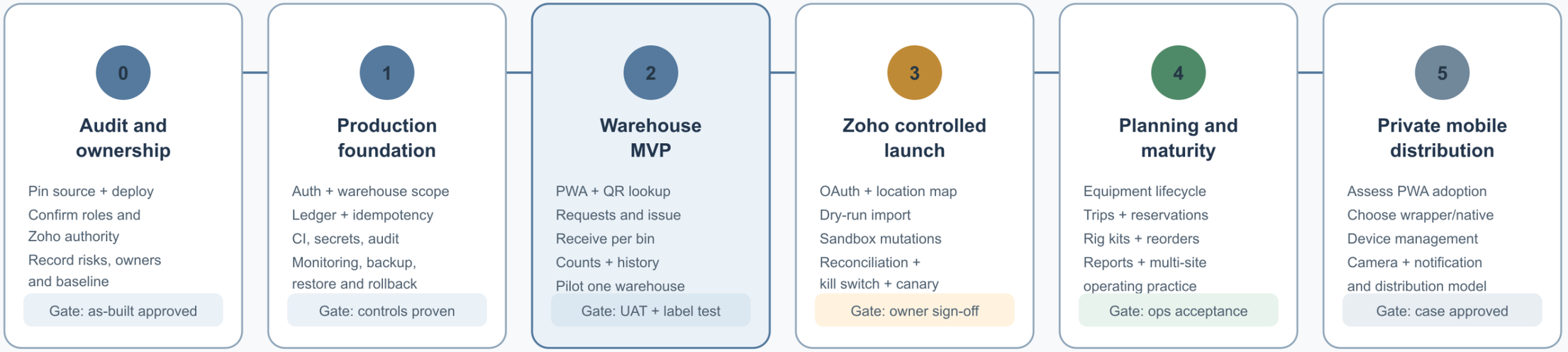
Movement 1→many SyncCommand attempts • Movement and business records 1→many AuditEvent • EquipmentAsset 1→many EquipmentLoan history

Unique product/bin balance • transactional movement + balance update • explicit units • append-only movements • linked corrections • scoped external IDs

Conceptual only: verify current schema, database capabilities, cardinalities and constraints against source before migration.

Production roadmap

Sequence by evidence and exit criteria • dates require source audit, ownership and capacity estimates



CROSS-PHASE RELEASE RULES

No live stock writes before:

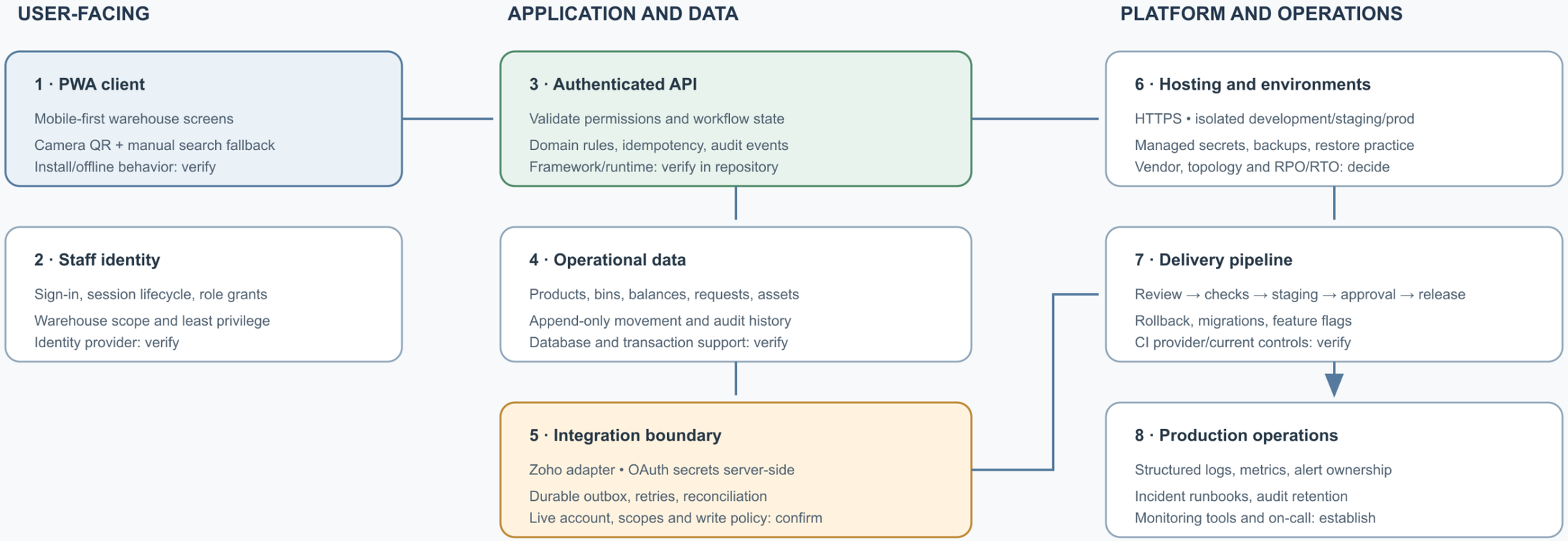
• source and current deployment are audited • authority and access policy are signed off • duplicate/concurrency tests pass • Zoho sandbox failures are rehearsed • restore, rollback and support ownership are proven

Phases can overlap only when their dependencies and exit evidence are satisfied; calendar estimates come after Phase 0.

This is a target delivery plan. Repository commit history reports some features already built; verify them before scheduling implementation work.

Production stack: target capabilities and decision status

PWA-first direction is agreed; exact frameworks, vendors and deployed controls require source/environment verification.



Technology selection and verification register

Agreed direction	PWA initial release • private company mobile app later • warehouse operations integrated with Zoho Inventory
Do not assume	Frontend/backend frameworks • database engine • identity provider • cloud host • CI vendor • current deployment status
Production gate	Inspect source and deployment configuration, confirm system-of-record policy, then record selected versions/vendors in ADRs.

Status is deliberately explicit: diagram describes a production-grade target, not evidence that these controls are currently deployed.